

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Smart Study Generator

Domain of Project – Education

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Drisya C C		ccdrisya28@gmail.com	8606303261

Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement -

Problem Statement: Smart Study Generator

The Challenge

Students often struggle to organize their studies, understand complex concepts, and prepare effectively for exams.

Learning resources are scattered across multiple platforms, making it difficult to access personalized guidance and practice materials.

The Objective

Develop an AI-powered Smart Study Generator that:

- Creates personalized study plans.
- Explains concepts in simple language.
- Generates quizzes for self-assessment.
- Provides study tips and learning support.
- Delivers an interactive AI chat experience using IBM watsonx.ai.

Proposed Solution -

Proposed Solution: AI-Powered Smart Study Generator

The proposed solution is an AI-powered Smart Study Generator developed using **IBM watsonx.ai, IBM BOB, Python Flask, and Granite Models**. It helps students plan their studies, understand topics, and improve learning through an interactive AI assistant.

Key Features

- **Study Planner:** Generates personalized study schedules.
- **Concept Assistant:** Explains topics with simple examples.
- **Quiz Generator:** Creates practice questions for revision.
- **Study Support:** Provides learning tips and productivity guidance.

The application features a responsive web interface where students can interact with the AI assistant. By leveraging IBM Granite models, the system delivers accurate, personalized, and real-time academic support, making learning more effective and engaging.

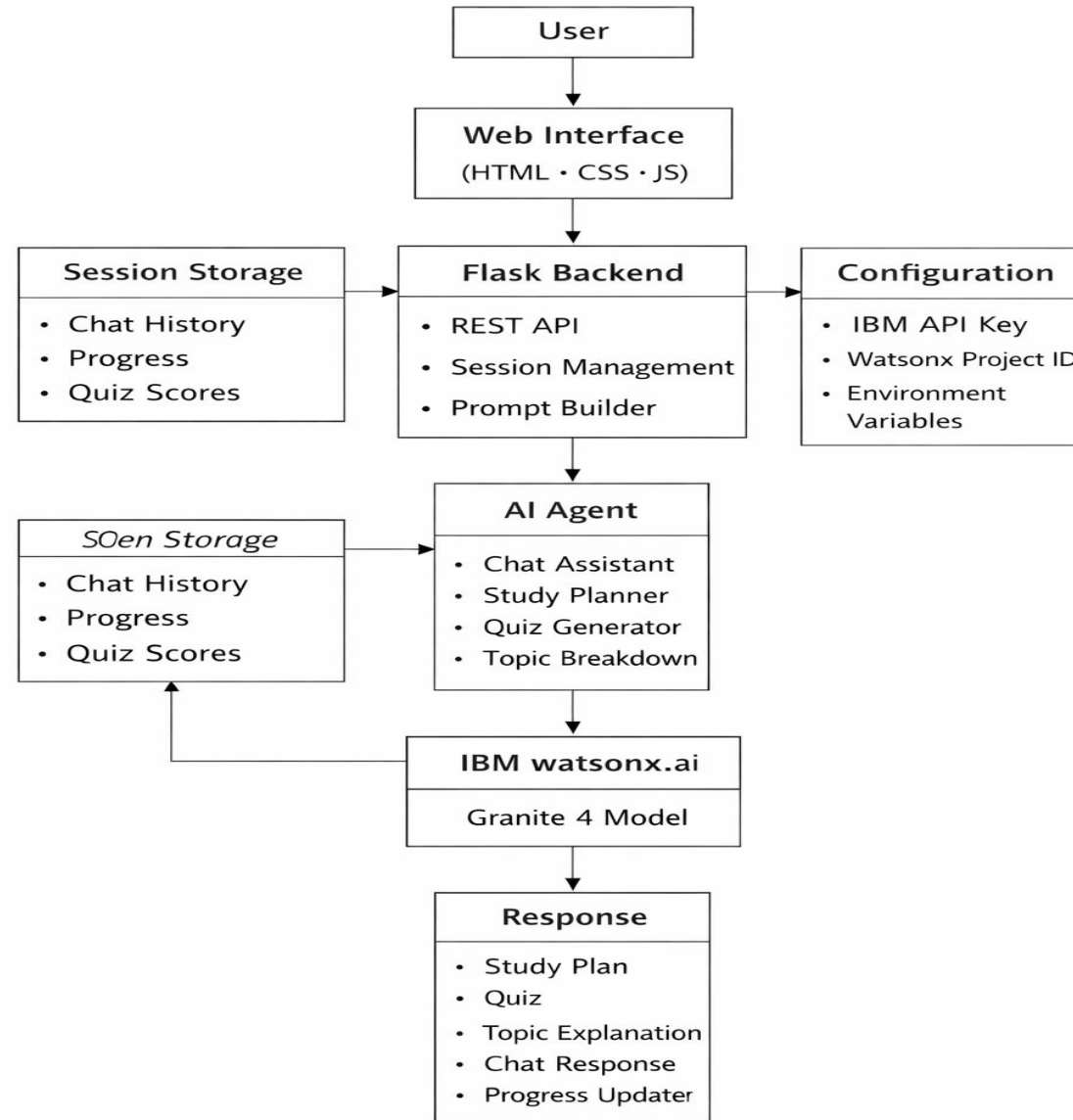
BOB component Used -

1. **IBM BOB** - Generates project code and accelerates application development using AI prompts.
2. **IBM watsonx.ai** - Provides access to IBM Granite foundation models for AI responses.
3. **IBM Granite Models** - Performs natural language understanding, reasoning, and personalized study assistance.
4. **IBM Cloud** - Hosts and manages watsonx.ai services securely.
5. **Python Flask** - Develops the backend web application.
6. **Bootstrap, HTML, CSS & JavaScript** - Builds the responsive user interface.

Technology Used

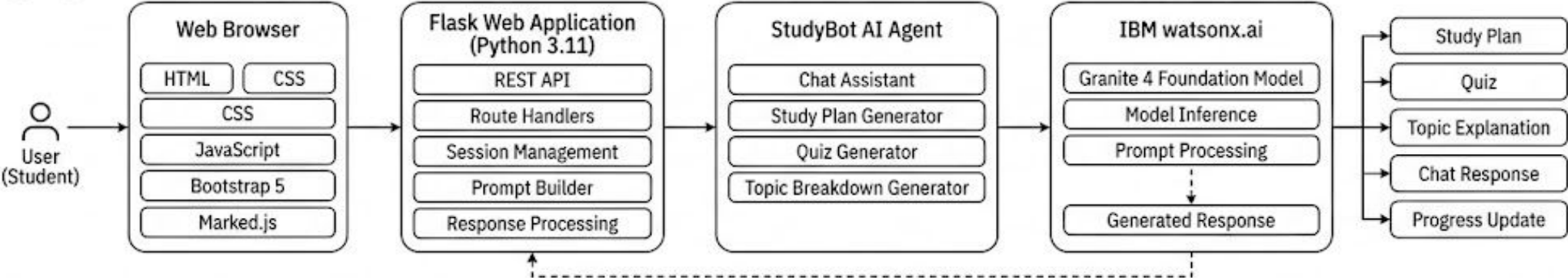
- **IBM Granite Model (Granite 3.3/3.2)** - Provides natural language understanding, reasoning, and personalized study assistance.
- **IBM watsonx.ai** - Enables access to Granite foundation models for AI-powered response generation.
- **IBM Cloud** - Provides a secure and scalable platform for hosting and integrating watsonx.ai services.
- **IBM BOB** - Assists in generating the application structure and accelerating AI-powered development.
- **Python Flask** - Used to build the backend and manage application logic, APIs, and user interactions.
- **HTML, CSS, JavaScript & Bootstrap** - Used to create a responsive and user-friendly web interface.

Block Diagram



Architecture Blueprint

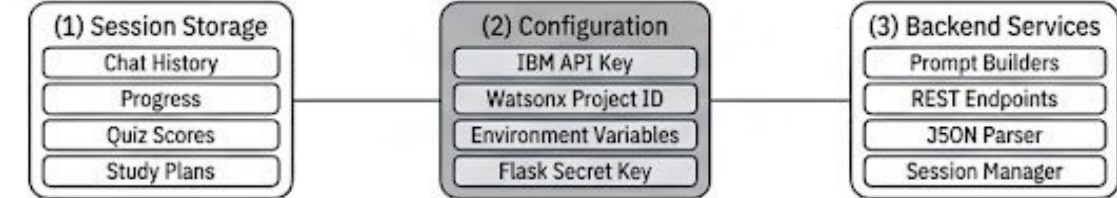
1 High-Level Architecture



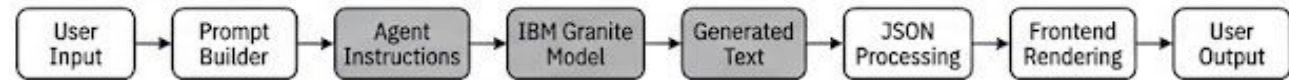
2 Request Processing Flow



3 Backend Components



4 AI Processing Pipeline



5 Technology Stack

Frontend	HTML5, CSS3, Bootstrap 5, JavaScript, Marked.js
Backend	Python 3.11, Flask, Flask-CORS
AI	IBM watsonx.ai, IBM Granite 4 (ibm/granite-4-h-small)
Configuration	python-dotenv
Deployment	Gunicorn, Docker, IBM Cloud Code Engine

Role of Agentic AI in the solution

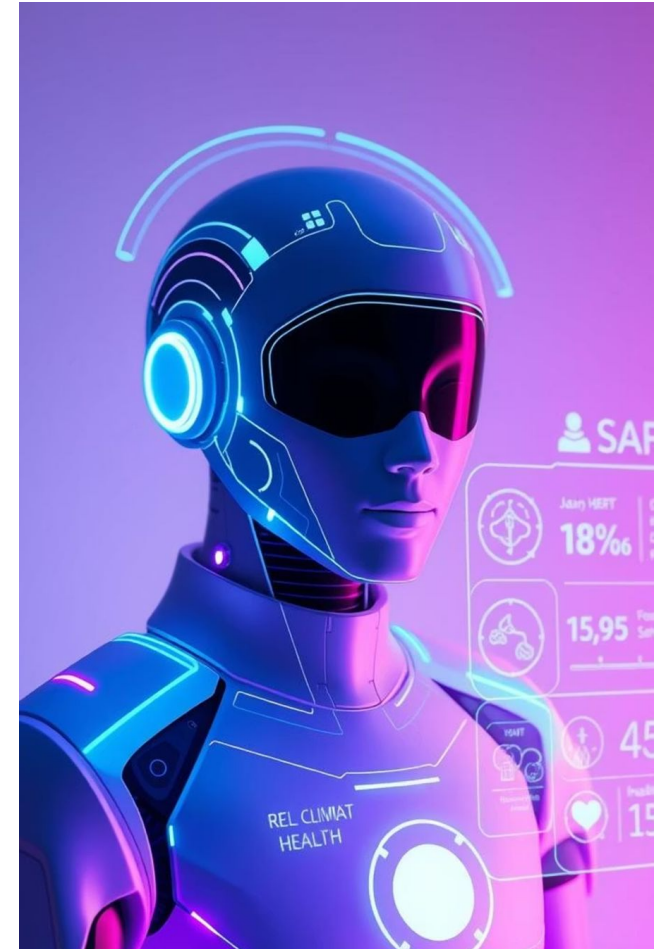
Role of Agentic AI in Smart Study Generator

Agentic AI enables the **Smart Study Generator** to function as an intelligent learning assistant rather than a simple chatbot. Powered by **IBM watsonx.ai** and the **IBM Granite model**, it autonomously understands user requests and generates personalized learning support.

The AI agent performs multiple tasks, including generating personalized study plans, answering academic questions, creating quizzes, and breaking down complex topics into easy-to-understand explanations. These capabilities are managed through task-specific prompts while maintaining a consistent learning experience.

The agent is context-aware, considering the student's selected subject, learning goals, available study time, and previous conversation history to deliver relevant and personalized responses. It also adapts dynamically based on user interactions, providing appropriate explanations and recommendations throughout the learning process.

Overall, Agentic AI makes the Smart Study Generator proactive, personalized, and goal-oriented, acting as an intelligent study companion that supports effective learning and continuous progress.



Novelty and Uniqueness

The proposed **Smart Study Generator** combines **Agentic AI**, **personalized learning**, and **IBM Granite models** to create an intelligent and adaptive study assistant.

- **Agentic AI-Based Learning** – Uses an AI agent that autonomously understands user requests and generates personalized study support instead of providing generic responses.
- **Personalized Study Experience** – Creates customized study plans based on the student's subject, learning goals, available study hours, and skill level.
- **Context-Aware Assistance** – Maintains conversation history and learning context to provide more relevant and coherent responses.
- **Multiple Learning Features** – Integrates study planning, topic explanations, quiz generation, and academic Q&A within a single platform.
- **Interactive Learning Support** – Provides instant quizzes, explanations, progress tracking, and Pomodoro-based study reminders to improve learning efficiency.
- **Powered by IBM watsonx.ai** – Built using IBM watsonx.ai and IBM Granite models for reliable, scalable, and intelligent content generation.

This makes the solution an intelligent, personalized, and goal-oriented AI study companion rather than a traditional study planner or chatbot.

Output

 **SmartStudy**

Chat

Study Plan

Quiz

Topic Breakdown

Progress

Select subject...

✦ Powered by IBM Watsonx.ai Granite

Your AI-Powered Study Companion

Personalised study plans · Smart quizzes · Topic breakdowns · Progress tracking

Start Learning ↓

Agent: **StudyBot** · Tone: *friendly and encouraging* · Speciality: *general academics* · Depth: *adaptive*

Chat with **StudyBot**

Clear

 Hi! I'm **StudyBot**, your AI study companion powered by IBM Watsonx.ai Granite.

I can help you with:

- 📅 **Study plans** tailored to your schedule
- 🧠 **Topic breakdowns** with examples and key concepts
- ✅ **Personalised quizzes** to test your knowledge
- 🕒 **Study-break suggestions** to keep you sharp

Select a subject in the top menu or just start chatting!

 **Subject Profile**

Active Subject

Choose subject...

Your Level

Intermediate

Learning Goal

e.g. Pass finals with A grade

Git Hub Link

<https://github.com/Ccdrisya/smart-study-generator.git>

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Future Scope

1. **AI-Powered Adaptive Learning**

The system can analyze students' learning progress, quiz performance, and study habits to automatically generate personalized study plans, recommend revision topics, and improve learning outcomes.

2. **Voice Assistant & Multilingual Support**

Future enhancements can include voice-based interaction and support for multiple languages, allowing students to communicate naturally and learn in their preferred language, making the platform more accessible and inclusive.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Drisya C C

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

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for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 07 Jul 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.